A FLAT FINITELY PRESENTED ALGEBRA WITH GEOMETRICALLY INTEGRAL FIBRES IS PROJECTIVE OVER THE BASE, WITHOUT NOETHERIAN HYPOTHESES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Approximation of the situation by one of finite type over the integers. More on Morphisms, labels more-morphisms-lemma-Noetherian-approximation-flat, more-morphisms-lemma-Noetherian-approximation-geometrically-reduced, more-morphisms-lemma-Noetherian-approximation-geometrically-irreducible and more-morphisms-lemma-Noetherian-approximation-combine (proof lines 2270-2273). Produces a cocartesian diagram with a base of finite type over the integers over which the same hypotheses hold, so that the Noetherian statement applies.
- The Noetherian version of the projectivity statement, and ascent of projectivity along a base change. Same chapter flat.tex, label lemma-fibres-irreducible-flat-projective (proof line 2275, claimed as S14-algebra-fibres-irreducible-projective); Algebra, label algebra-lemma-ascend-properties-modules (proof line 2278). The first gives projectivity over the approximating base and the second ascends projectivity to the given base.

Source reference. Original lemma 05A3: sources/stacks-chow/tags.txt, source lines 6836–6836. Navigation pointer for \ref{algebra-lemma-ascend-properties-modules} used inside the counted span (line 2278); the argument that consumes it is retained in the span.

Source reference. Original lemma 05FS: sources/stacks-flat/pinned-original-source.tex, source lines 2179–2179. Navigation pointer for \ref{lemma-fibres-irreducible-flat-projective} used inside the counted span (line 2275); the argument that consumes it is retained in the span.

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Source reference. Original lemma 05FL: sources/stacks-chow/tags.txt, source lines 7030–7030. Navigation pointer for \ref{more-morphisms-lemma-Noetherian-approximation-combine} used inside the counted span (line 2273); the argument that consumes it is retained in the span.

Source reference. Original lemma 05FE: sources/stacks-chow/tags.txt, source lines 7023–7023. Navigation pointer for \ref{more-morphisms-lemma-Noetherian-approximation-flat} used inside the counted span (line 2270); the argument that consumes it is retained in the span.

Source reference. Original lemma 05FH: sources/stacks-chow/tags.txt, source lines 7026–7026. Navigation pointer for \ref{more-morphisms-lemma-Noetherian-approximation-geometrically-irreducible} used inside the counted span (line 2272); the argument that consumes it is retained in the span.

Source reference. Original lemma 05FG: sources/stacks-chow/tags.txt, source lines 7025–7025. Navigation pointer for \ref{more-morphisms-lemma-Noetherian-approximation-geometrically-reduced} used inside the counted span (line 2271); the argument that consumes it is retained in the span.

Lemma 0.1. *Let R be a ring. Let $R \to S$ be a ring map. Assume*

(1) *$R \to S$ is of finite presentation and flat, and*
(2) *every fibre ring $S \otimes_R \kappa(\mathfrak{p})$ is geometrically integral over $\kappa(\mathfrak{p})$.*

Then S is projective as an R-module.

Proof. We can find a cocartesian diagram of rings

$$\begin{array}{ccc} S_0 & \longrightarrow & S \\ \uparrow & & \uparrow \\ R_0 & \longrightarrow & R \end{array}$$

such that R_0 is of finite type over $\mathbf{Z}$, the map $R_0 \to S_0$ is of finite type and flat with geometrically integral fibres, see More on Morphisms, Lemmas 05FE, 05FG, 05FH, and 05FL. By Lemma 05FS we see that S_0 is a projective R_0-module. Hence $S = S_0 \otimes_{R_0} R$ is a projective R-module, see Algebra, Lemma 05A3. □